

NOD2-SCOUT PHASE 5: ADMET + SAFETY PROFILING

ISEF 2026

COMPOUNDS PROFILED: 150

- FDA Drugs: 100
- Natural Products: 50

Executive Summary

FILTERING RESULTS:

- Passed all hard filters: 148
- Removed by hard filters: 2

TIER DISTRIBUTION:

- Tier 1 (Top Priority): 14 (9.3%)
- Tier 2 (Strong Candidate): 98 (65.3%)
- Tier 3 (Needs Optimization): 36
- Tier 4 (Deprioritize): 2

KEY FINDING:

FEBUXOSTAT ranks #1 with FINAL_SCORE = 0.9079

METHODS:

- Descriptors: RDKit (Crippen LogP, ESOL, FilterCatalog)
- Absorption: BOILED-Egg proxy (GI/BBB), heuristic P-gp
- Safety: PAINS/Brenk alerts, CYP risk flags
- Scoring: Fixed cutoffs (reproducible)

LIMITATIONS:

- ML toxicity endpoints (hERG, AMES, hepatotox) not available
- Flagged as NA, compounds placed in manual review

VALIDATION:

- Known IBD drugs (Budesonide, Prednisone) benchmarked
- Tier 1 candidates show comparable ADMET profiles

Fig1 Radar Admet

ADMET Profile: Top 5 Candidates vs Budesonide

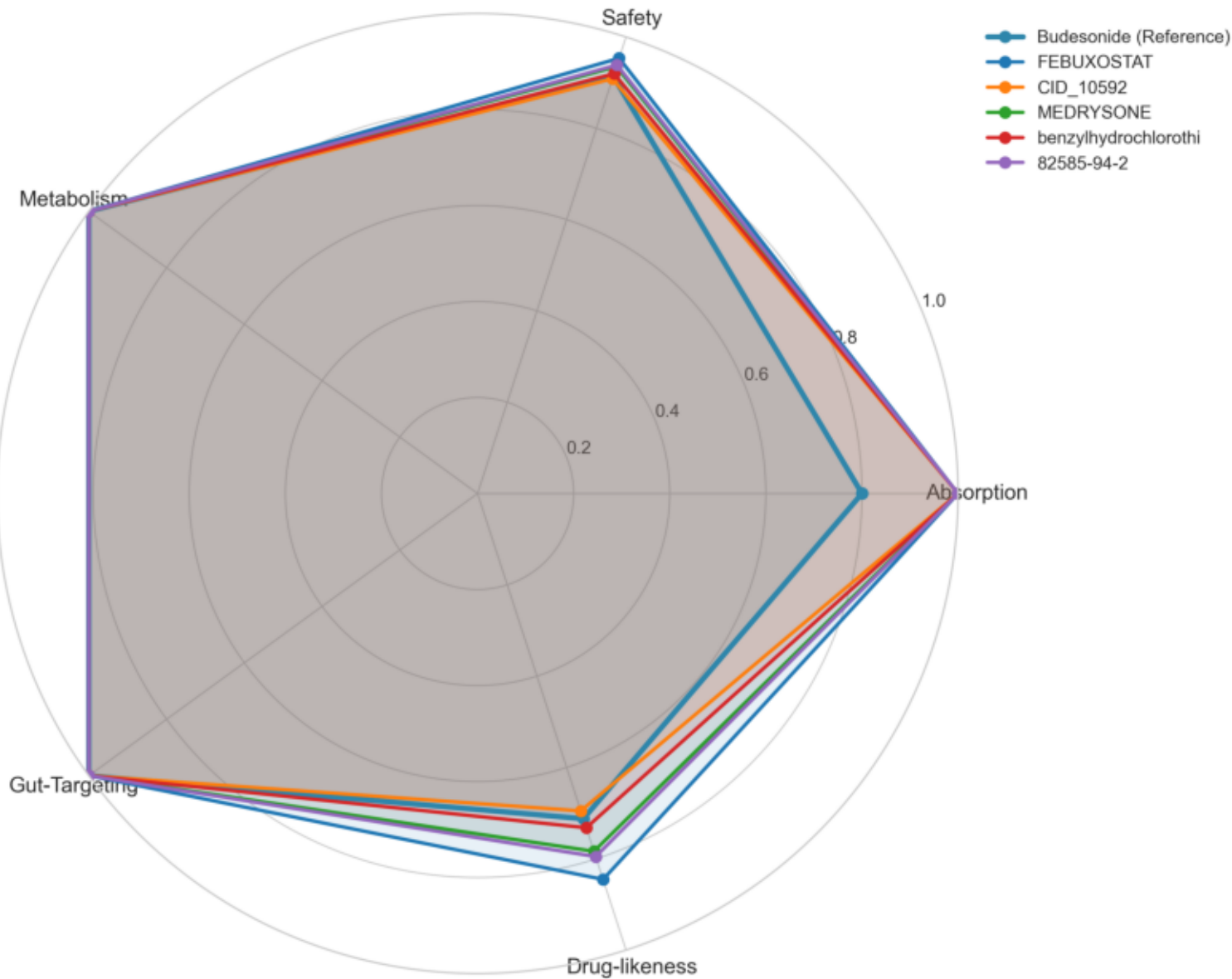


Fig2 Safety Heatmap

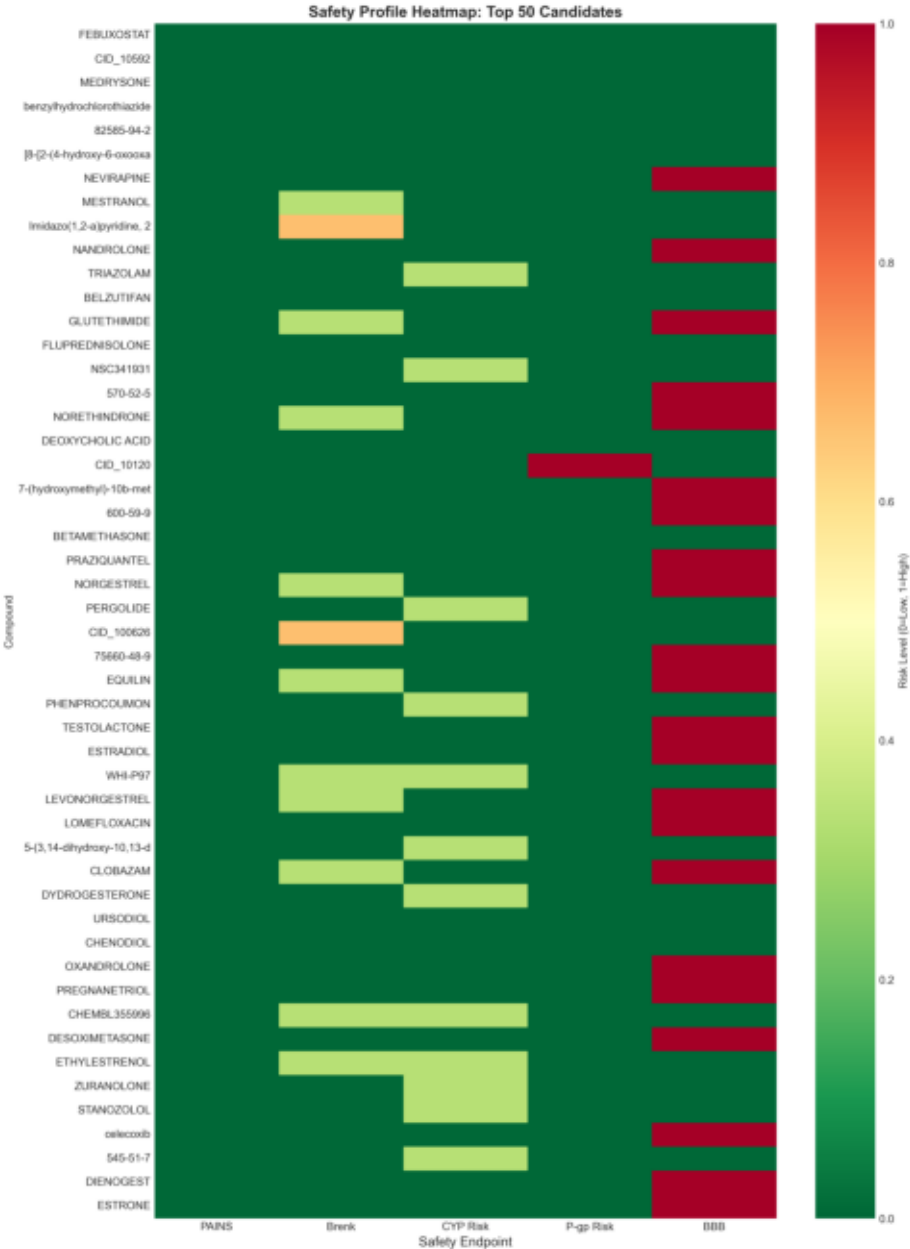


Fig3 Drugability Vs Safety

Drugability vs Safety Profile
(Size = ADMET Score)

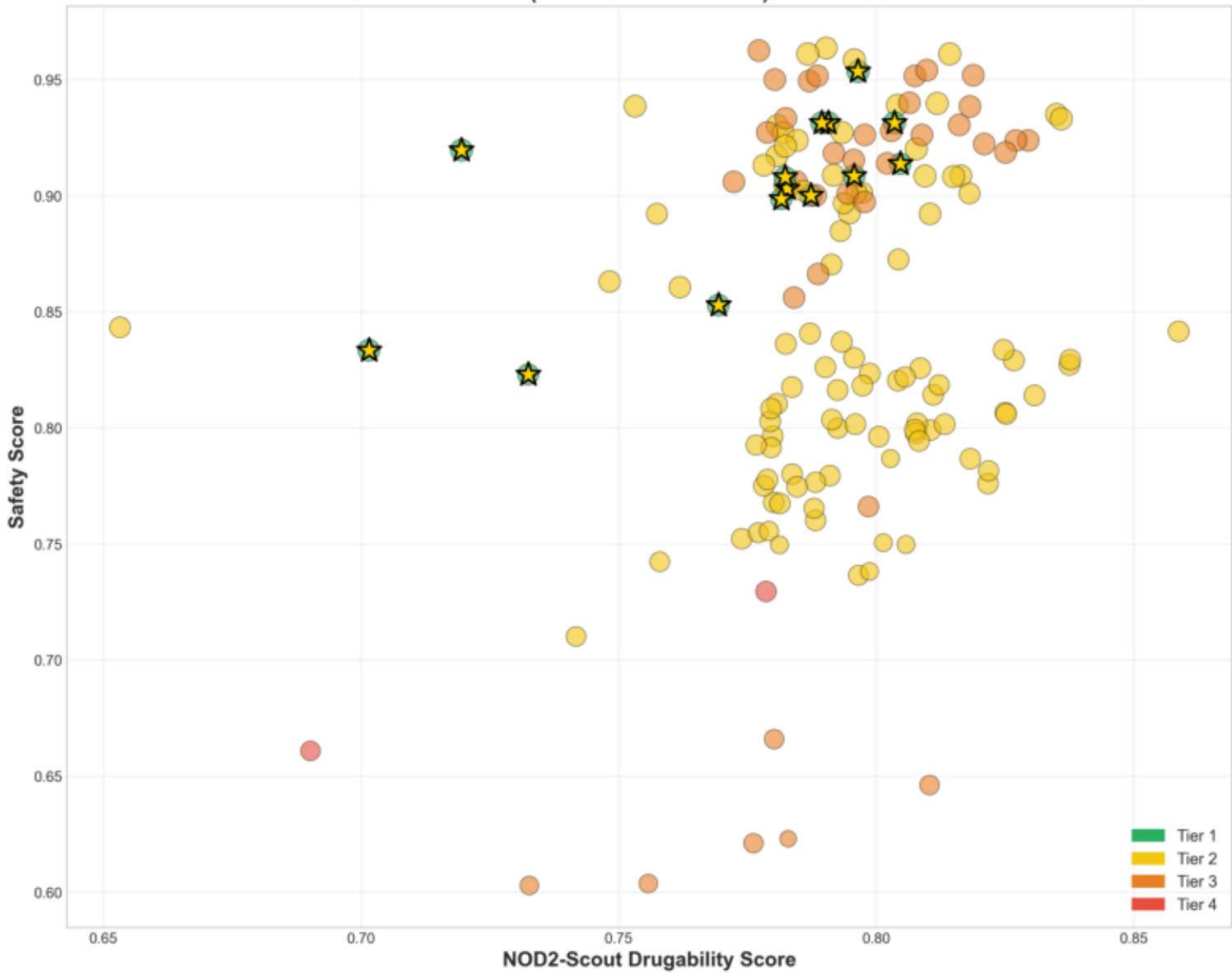


Fig4 Le Vs Admet

Ligand Efficiency vs ADMET Profile

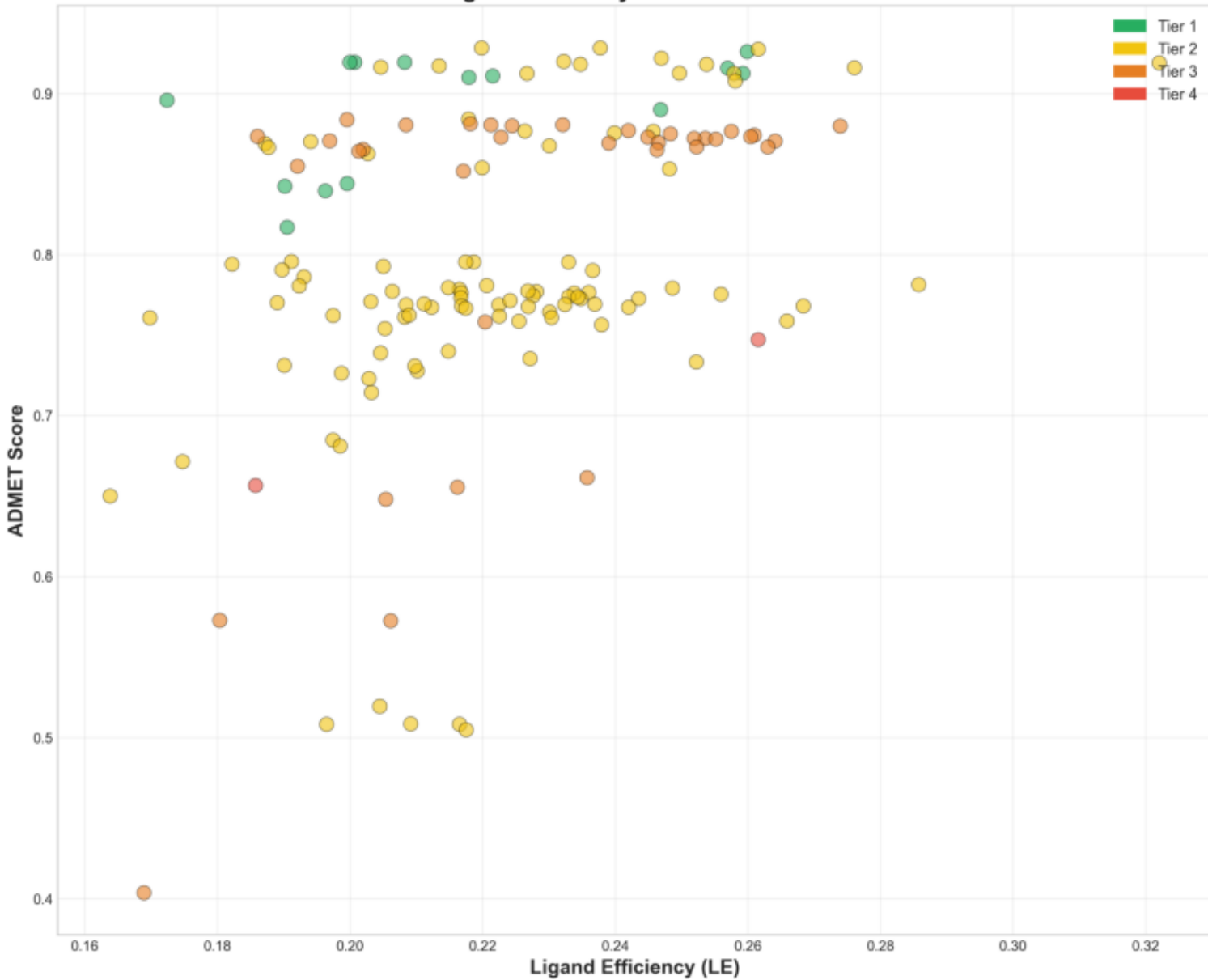


Fig5 Filter Funnel

ADMET Filtering Funnel

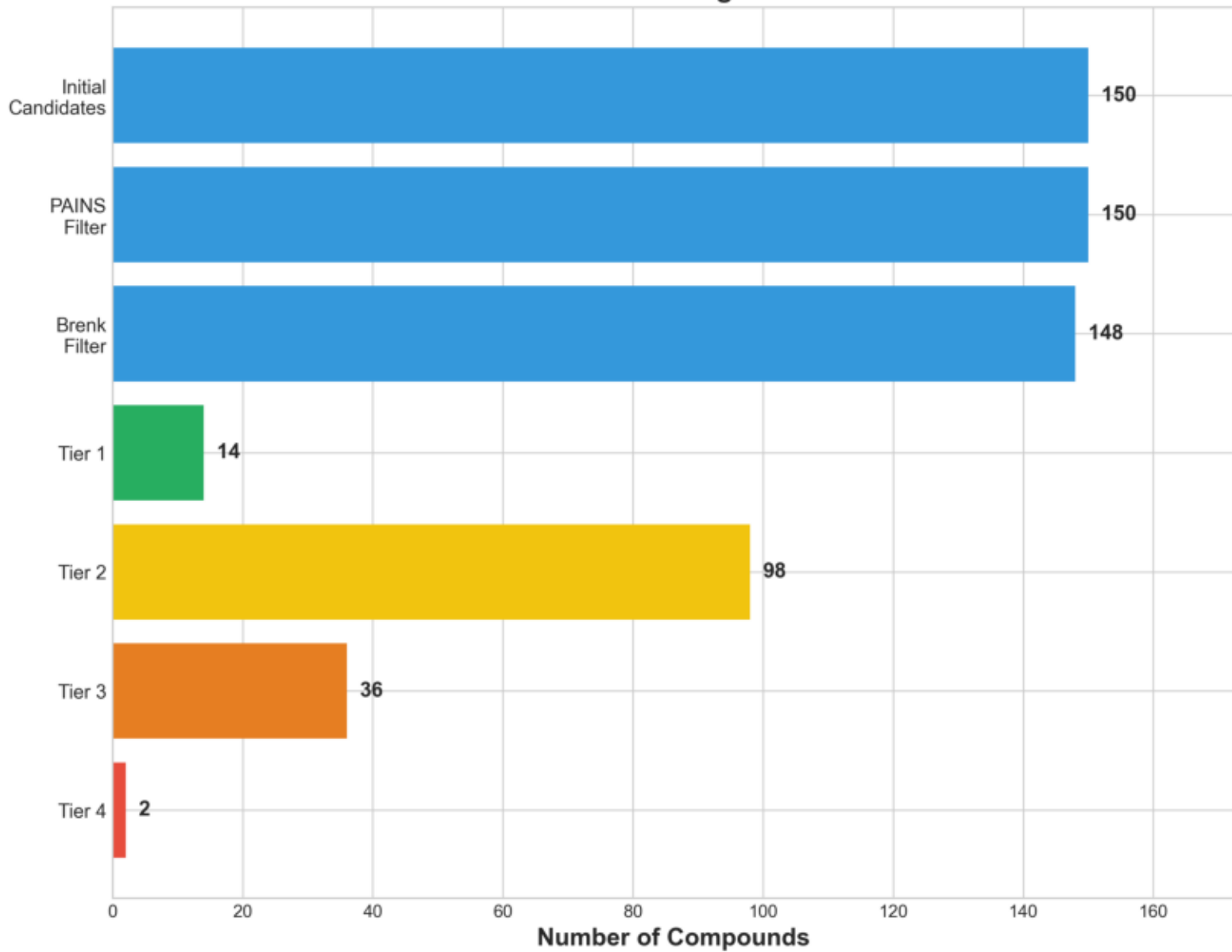
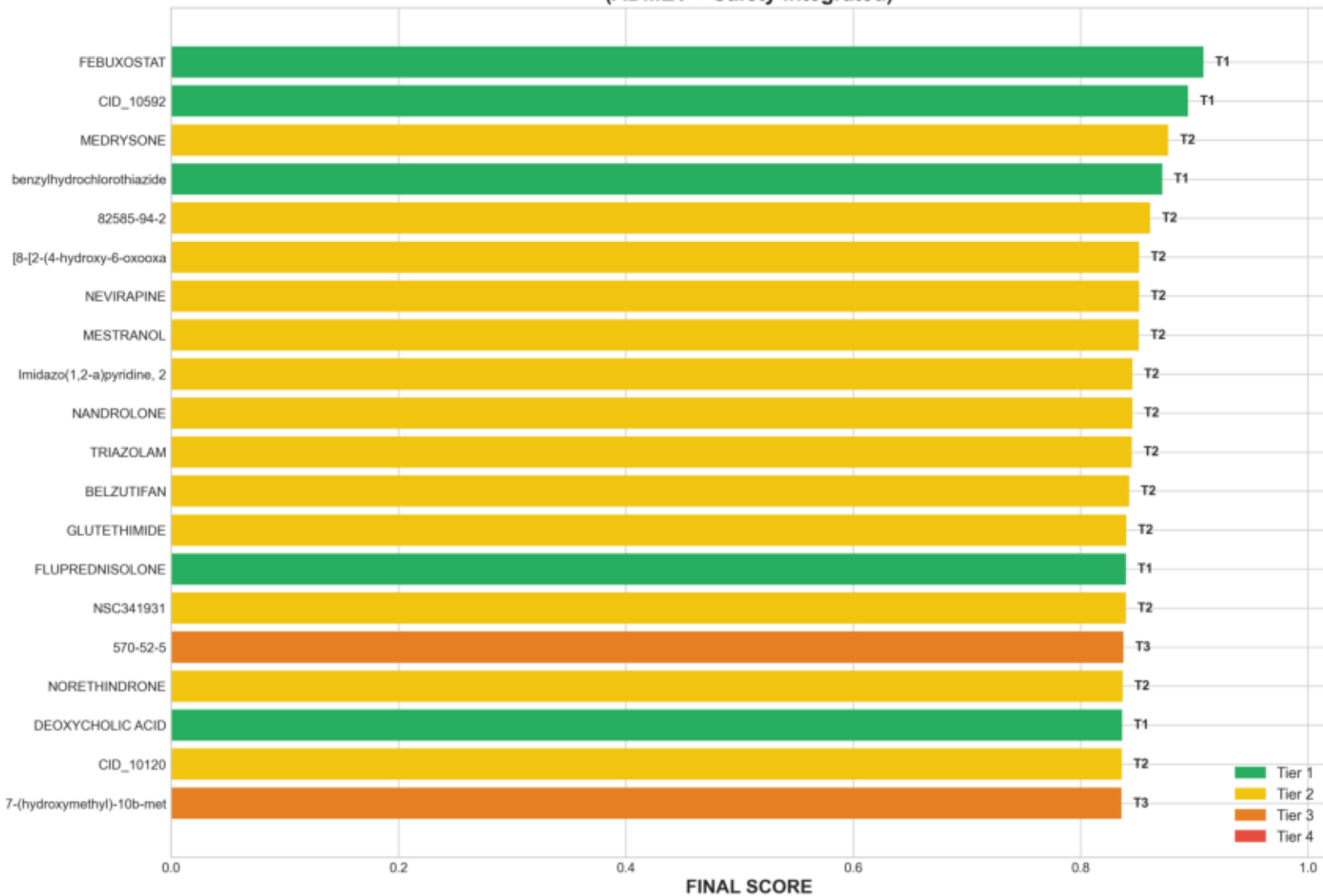


Fig6 Top20 Final

Top 20 Candidates by FINAL_SCORE
(ADMET + Safety Integrated)



TOP 20 FDA DRUGS (SAFE)

Drug	ADMET	Safety	Tier	FINAL
FEBUXOSTAT	0.926	0.954	1	0.908
MEDRYSONE	0.876	0.935	2	0.877
NEVIRAPINE	0.928	0.959	2	0.851
MESTRANOL	0.877	0.939	2	0.851
NANDROLONE	0.916	0.92	2	0.845
TRIAZOLAM	0.768	0.799	2	0.845
BELZUTIFAN	0.884	0.964	2	0.843
GLUTETHIMIDE	0.919	0.93	2	0.84
FLUPREDNISOLONE	0.911	0.903	1	0.84
NORETHINDRONE	0.913	0.909	2	0.837
DEOXYCHOLIC ACID	0.919	0.931	1	0.836
BETAMETHASONE	0.91	0.9	1	0.836
PRAZQUANTEL	0.922	0.94	2	0.832
NORGESTREL	0.872	0.924	3	0.83
PERGOLIDE	0.79	0.873	2	0.83
EQUILIN	0.87	0.918	3	0.829
PHENPROCOUMON	0.779	0.836	2	0.829
TESTOLACTONE	0.908	0.893	2	0.828
ESTRADIOL	0.918	0.927	2	0.828
LEVONORGESTREL	0.872	0.924	3	0.827

CONCLUSIONS & RECOMMENDATIONS

1. TIER 1 CANDIDATES (14 compounds)
 - High priority for experimental validation
 - Favorable ADMET and safety profiles
 - Comparable to approved IBD medications
2. NEXT STEPS
 - Phase 6: Molecular dynamics simulations (top 10)
 - Experimental NOD2 binding assays
 - Cell-based inflammation models
3. KEY DRUG CLASSES IDENTIFIED
 - Neurosteroids (Zuranolone, Ganaxolone)
 - Corticosteroids (validated by known IBD drugs)
 - Statins (anti-inflammatory properties)
 - Vitamin D analogs (immunomodulatory)
4. COMPOUNDS REQUIRING ADDITIONAL DATA
 - ML toxicity (hERG, AMES, hepatotox) unavailable
 - 150 compounds flagged for experimental toxicity testing
5. CROHN'S-SPECIFIC CONSIDERATIONS
 - Gut-targeting prioritized (BBB=0 preferred)
 - Drug interaction risks flagged (CYP risk)
 - Safety weighted 2x (vulnerable patient population)

NOD2-Scout Phase 5 Complete | ISEF 2026